

Physical and Social Adaptation for Assistive Robot Interactions

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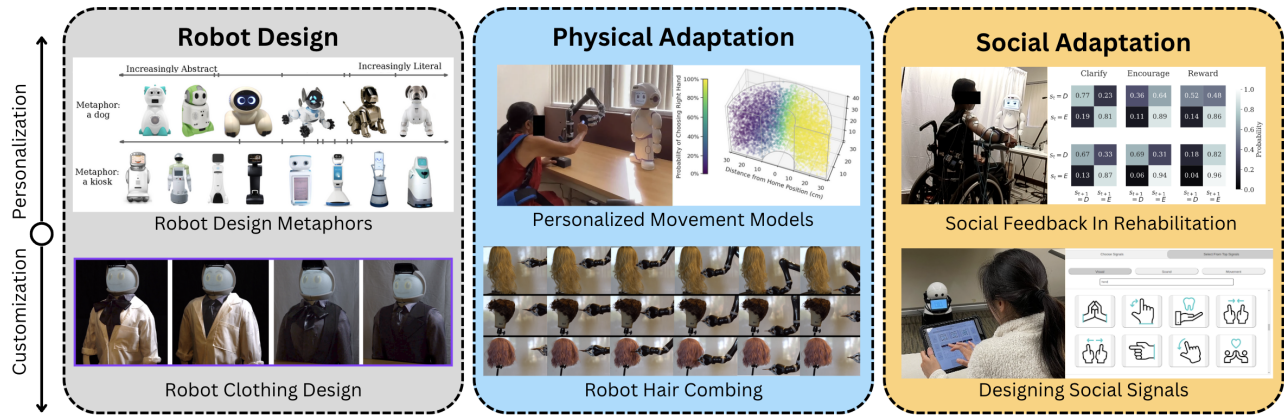


Figure 1: Overview of the ways assistive robots adapt to users.

Abstract

Robots have the potential to provide users with limited mobility additional ways of interacting with the world around them. However, each user has preferences for how they interact with these physical interfaces. My dissertation research develops tools and algorithms to allow robot interactions to adapt to the individual needs of users. In particular, I develop ways to adapt a robot's design, physical movements, and social behaviors. By adapting robots to users I hope to develop systems that more holistically aid users with limited mobility.

CCS Concepts

• **Human-centered computing** → **Systems and tools for interaction design**; **User interface programming**.

Keywords

Human-Robot Interaction, Assistive Robotics, Personalization, Customization

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1 Introduction

There are currently 2.4 billion people worldwide who are living with a health condition that can benefit from physical therapy [1]. However, rehabilitation therapists have limited time with patients to provide one-on-one attention. Thus the success of rehabilitation relies on a patient's continued and focused independent practice [5]. A major barrier to completing rehabilitation exercises is a lack of patient motivation [31], and if the rehabilitation process is not followed, functional decline can lead to a lower quality of life [36, 47]. Robots have the potential to be influential in rehabilitation by providing physical support [38] and social assistance [41]. Robots can tirelessly support users without them feeling judged [24], and different system behaviors can lead to increased use of the system and result in more effective rehabilitation [44, 48].

Prior work has identified several personal and contextual factors that contribute to actual system use [2], and the way that robots should respond to these factors can result in different and sometimes opposing behaviors for each individual user. In order for robots to be useful to individuals, they must adapt to their needs and preferences. **My dissertation research focuses on how users can adapt generic robot systems to their specific needs.** In particular, I investigate three area of robot adaptation: robot appearance, physical adaptation, and social adaptation. Across these three areas, I emphasize the importance of two kinds of adaptation: *personalization*, where robots adapt automatically through interaction, and *customization*, where users explicitly modify the robot and its behaviors. Personalization requires little work from

the user and can improve user experiences but can result in incorrect user assumptions, whereas customization provides the user with autonomy but may require additional effort from the user. My volunteer work with Queer in AI [39] and Queer in Robotics [28] has particularly identified that user's autonomy is a necessary component of AI systems, especially for systems interacting with marginalized populations [16].

2 Robot Appearance

Robots are constrained by their embodiment—e.g., a robot without a mechanism for locomotion will not be able to navigate indoors. The same is true of a robot's social attributes. In order for robots to successfully adapt, their embodiment must align with their intended application. I have explored how embodiment affects expectations of robots [17], and how we may *customize* these expectations through clothing [13].

2.1 Expectations of Robot Embodiment

Research in several applications of human-robot interaction (HRI) has found that the physical design of a robot affects how users expect to interact with the robot [9, 10, 27, 29], and performing to these expectations is a major factor that drives adoption [7, 22]. However, there was no principled way to link robot designs to expectations before using them to develop assistive applications.

I applied the idea of *design metaphors* [34] from interaction design to understand how embodiment affects interaction expectations. Design metaphors describe novel technology in terms of something familiar as a didactic tool; for example, a computer desktop mimics a physical desktop by allowing you to rearrange, sort, and move files. I collected a dataset of 165 different *socially interactive* [20] robot embodiments and the design metaphors that people associated with these robots [17]. I additionally collected social expectations using the RoSAS scale [6], and functional expectations using the EmCorp scale [23] for each of these robots from 1,608 people through an online survey. An example robot from our dataset is shown in Figure 2, along with its design metaphors, social expectations, and functional expectations.

We found that the metaphors used to describe the robots influenced how people expected to interact with the robot. We produced data visualizations to interact with this dataset and described a process that robot study designers can use to select a robot for a specific task to match robot expectations to the task. In follow-up work we also showed that the images of the robots could be used to predict social and functional expectations for unseen robots [14] using Large Vision-Language Models. This dataset also provides a resource for future research in the interplay between language, robot design, and user expectations.

2.2 Modulating Expectations through Clothing

While understanding a robot's embodiment is useful to align expectations and task, buying a new robot for a particular task can be prohibitively expensive for users. I identified that clothing can be a low-cost alternative to modify a robot's appearance to align with the robot's intended use [13]. Sewing can also be a way to practice fine motor skills [35], and involving the user in customizing a robot's appearance can lead to increased use of the robot [43].

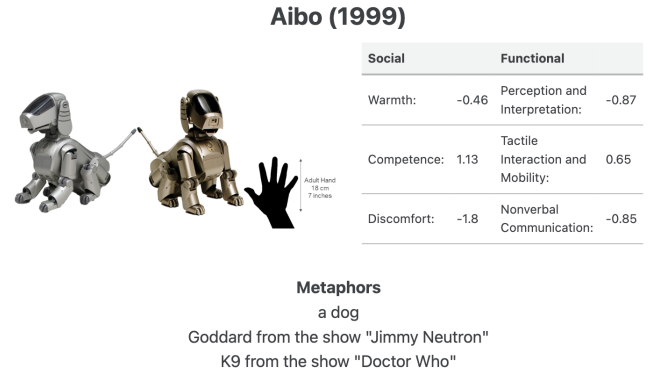


Figure 2: Examples datapoint from the MUFaSAA Dataset.

To explore this, I developed a design process to modify a robot's appearance through clothing by using the elements of fashion to reflect different characteristics of the clothing. In this case, the clothing reflected different perceived genders for a robot working in jobs of different social roles. Participants rated perceived gender, social attributes, and anticipated role of the robot. We found that the clothes the robot wears can significantly affect the robot's perceived gender and expectations of the task the robot is designed for compared to the unclothed robot. In addition to these benefits, participants did not perceive social differences based on the perceived gender of the robot. This allows designers to develop robots with different social identities without damaging the user's expectations that the robot can do a particular task. We believe that using this process, designers can be deliberate of the social identity that they design for a robot in order to subvert harmful human stereotypes of tasks, which has previously been found to increase participation in education [46].

3 Physical Adaptation

Robots can physically interact with the world, providing users with limited mobility an additional interface to interact with their surroundings. To adapt how robots move, I explored how robots can learn *personalized* models of the user [11], and how users can *customize* how a robot engages with them in personal care [18].

3.1 Developing Personalized Movement Models

In rehabilitation contexts, adapting the expected physical exertion of the task is important to make the task challenging but not too difficult [37]. To address this, I developed the Bimanual Arm Reaching Test with a Robot (BARTR) for post-stroke participants in collaboration with physical therapists [11]. In this task, the QTRobot provided instructions for the task, and a Kinova JACO2 arm [26] moved a target to different points in the space in front of the participant. Each participant attended 3 sessions and in each session, they performed 100 reaches using either hand to reach toward the target and 100 reaches using only their more-affected side to reach toward the target. In total, I recruited 14 post-stroke participants and 10 baseline neurotypical participants to complete the study.

Using each participant's data, I developed personalized models for estimating the time to reach the target for each hand, and a model to predict the hand that the participant selected to perform

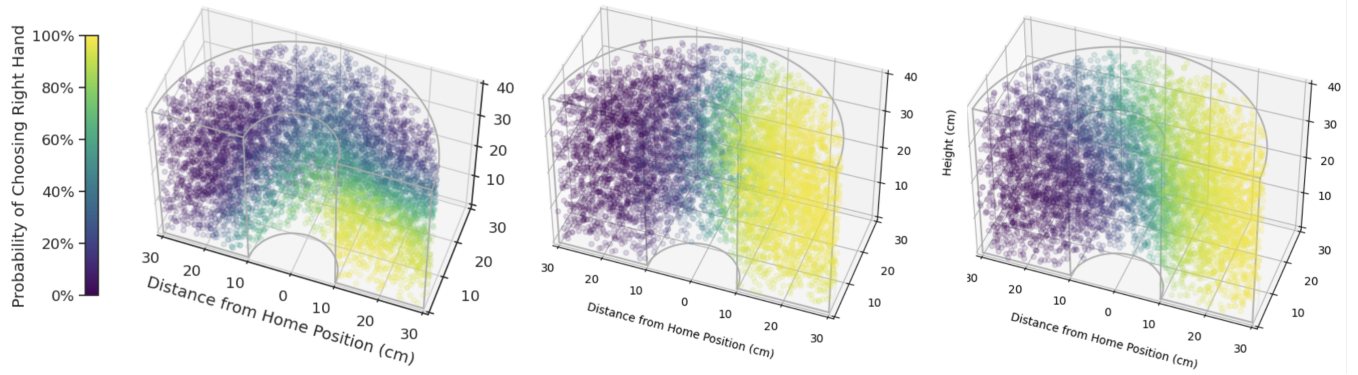


Figure 3: Personalized models of arm use. Based on users interactions, we identified where post-stroke participants over-relied on their less affected side. These visualizations can allow both neurorehabilitation specialists and post-stroke participants understand their personal movement patterns.

the reaching task. Traditionally, arm function is evaluated by a neurorehabilitation specialist, and only measures arm function in isolation [21]. By developing this system, I enabled automatic data collection and evaluation of arm function in a realistic task that can be used to monitor rehabilitation progress.

These personalized models revealed individual differences in arm movements, shown in Figure 3. I created an algorithm to estimate a difficult-to-collect clinical metric for arm performance using these models called *nonuse*. *Arm nonuse* [3] is defined as the difference between a post-stroke participant’s performance in conscious practice and performance in ecological use [3, 45]. This metric encapsulates the goal of physical therapy to transfer from the clinic to daily life. The clinical standard to assess this metric is only valid once because it requires the participant to be unaware of the test [42]. My algorithm to calculate *nonuse* from BART is highly correlated with the clinical standard for *nonuse* while also maintaining high test-retest reliability. This work highlighted that designing assistive systems to be adaptive serves a dual purpose of both assisting the user and providing information that can be leveraged by professionals making decisions.

My followup work has also identified how the data collected through this interaction can be used to identify personalized regions of difficulty [15]. I additionally advised an undergraduate student to develop a method to automatically adapt rehabilitation task difficulty to the user to provide a challenge while accounting for fatigue [8].

3.2 Developing Haircare Robots

In addition to physical therapy exercises and monitoring functional gains, robots provide an interface that users with limited mobility can use for personal care. Through interviews with people in post-stroke care facilities, we found that there was a lack of robots designed for hair care. Hair can be a strong aspect of someone’s identity [32], yet people with limited mobility may not have access to caregivers to do this task [30]. By providing users with a robot to accomplish hair combing, we aim to allow users with limited mobility a way to help re-establish their identity themselves.

Manipulating arbitrary non-rigid objects such as hair is a notoriously unsolved problem in robotics (e.g., styling the hair), so I

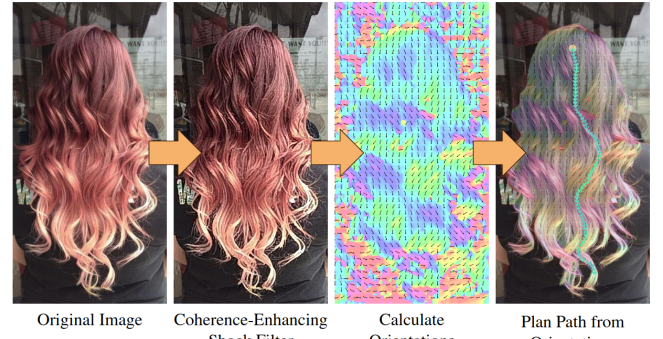


Figure 4: The algorithm we developed allows users to comb to generate paths through hair in a single click.

framed the problem as detangling unstyled hair. I developed a system that used a Kinova JACO2 arm [26] and a kinect depth sensor to comb a user’s hair. We selected this arm because it is designed for users with limited mobility, so it can serve other purposes for the user in addition to hair combing. The other benefit of using a general purpose arm is that users can customize the comb that the robot uses to comb their hair. I developed a path planning algorithm (shown in Figure 4) that generates a robot combing trajectory that follows the flow of the user’s hair from a single click. The arm followed this path while exerting minimal force on the user. Users can click the part of the hair that the robot should comb, allowing them to customize the way that the robot combs their hair.

I performed an initial online study to evaluate the hair combing system. I found that the path planning algorithm generated paths that were as good as human-drawn paths, and participants found that the hair looked considerably neater following the robot interaction. I additionally identified 3 factors to evaluate hair combing systems: effectiveness of haircombing, use of force, and quality of robot movement. These factors allow future iterations of this project with users with limited mobility to be evaluated.

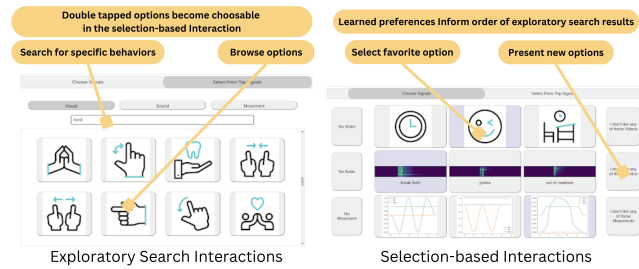


Figure 5: The RoSiD interface allows users to engage in exploratory search as well as traditional preference learning while designing robot behaviors.

4 Behavioral Adaptation

In order to adapt the social and communicative aspects of assistive robots, I explored how robots can automatically *personalize* their behaviors to users through interaction [19] and how users can *customize* robot signals through explicitly selecting signals aligned with their preferences [12].

4.1 Feedback Preferences in Rehabilitation

Previous works have shown that different kinds of social feedback in rehabilitation exercises can encourage users to better adhere to physical therapy regimens. I focused on social personalization for a robot that helps children with cerebral palsy become acclimated to orthosis use [19]. The goal of using an orthosis is to reduce contracture over longer time horizons [25]. Adherence is necessary for functional improvement, but children do not always respond well to traditional therapy and can benefit more from gamified interactions [4]. I designed an interaction where a robot engaged users in a one-on-one number guessing game to repeat pronation and supination; we developed an interface that allowed the participant to communicate with the robot through thumbs up and thumbs down gestures. The robot provided social feedback in the form of a rewarding, encouraging, or clarifying statement, which have been used in other robot coaching tasks [40, 44].

The study team recruited 10 participants with cerebral palsy to participate in the study, and I developed personalized models for how the robot’s choice of feedback affected the engagement of the participant in the game. I found that the participants’ dynamics of engagement were represented by different clusters. Using leave-one-out cross-validation, I found that these clusters described the unseen user. In addition, by using these clusters, an algorithm can select robot actions that significantly increase the expected engagement level of the participant by approximately 5% using simulated data from the participants’ actual interaction.

4.2 Designing Social Signals

In addition to social feedback to encourage the user, robots must also communicate important aspects of their status with the user. The behaviors that robots use to signal their status depend on many personal, social, and cultural factors. Since these factors are unknown before these robots are deployed to assist people, they must be customizable to suit the preferences of individual users. While computers can often successfully utilize common signal metaphors,

robot signals are cannot always use these metaphors due to the added social modalities of robots embodiments [10].

In robotics research, preference learning allows users to specify how robots should perform complex behaviors by selecting their favorite behavior from a small set of behaviors. As users repeatedly select robot behaviors, the robot learns the optimal behaviors for a particular user. However, I identified that these methods are structured around a fundamental assumption that the user understands what the robot is capable of doing, however most end-users will be jointly learning about what the robot can do to decide what they would like it to do. I developed an interface for teaching robots that allows users to engage in *exploratory search* [33] to better learn what the robot is capable of while they specify their preference. The Robot Signal Design (RoSiD) interface (shown in Figure 5) allows users to browse, filter, and select from a large library of robot behaviors, while also providing the traditional preference learning interaction [12].

I evaluated the RoSiD interface with 25 participants that engaged in designing signals for an assistive item-finding robot. We used a Mayfield Kuri robot equipped with a screen, and each signal consisted of a video, audio, and gestural component. Participants designed signals to represent that the robot was idling, searching for an item, found an item, or had information about an item’s location. Users reported that the RoSiD interface was easy to use, quick to learn, and allowed them to select the signals that they liked. Our post-hoc analysis showed that using data collected from this interface can also be used to learn representations that allow users to obtain their desired signals more quickly [15]

5 Conclusion and Future Work

In my dissertation, I investigated how assistive robots functioning as interfaces with the physical world can adapt their appearance, physical interaction, and social interaction across several assistive domains. I emphasize the importance of both automatically *personalizing* and manually *customizing* robots in each of these domains.

My short-term future directions are to develop tools to allow non-expert users to design and fabricate their own cloth-based robot clothes to modify a robot’s embodiment, particularly for low-cost robots such as the Blossom robot to provide more accessible assistive robots to users with limited mobility. Allowing users to create their own clothes for the robot to modify its appearance can quickly adapt the robot’s embodiment to the user’s preferences. The manufacturing process can also function as a way to practice fine motor manipulation skills for some users.

My long-term goals aim to incorporate all three types of adaptation into a single accessible and long-term robot interaction for rehabilitation for post-stroke users. Robots for rehabilitation have focused on either social or physical adaptation. While my work with post-stroke participants has shown that robots can provide social and physical assistance, the socially assistive robot did not adapt its social behaviors to the individual user. I believe that social and physical assistance are deeply intertwined—for example, users that are becoming unmotivated to perform repeated exercises may benefit from either social or physical adjustments of the system. By providing holistic interactions, I hope to provide interactions that meaningfully assist users with limited mobility.

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